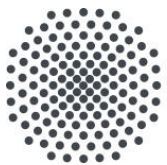


Collaborative Model-Driven Quantum Software Engineering: A Hands-On Tutorial with Qonstruct



University of Stuttgart

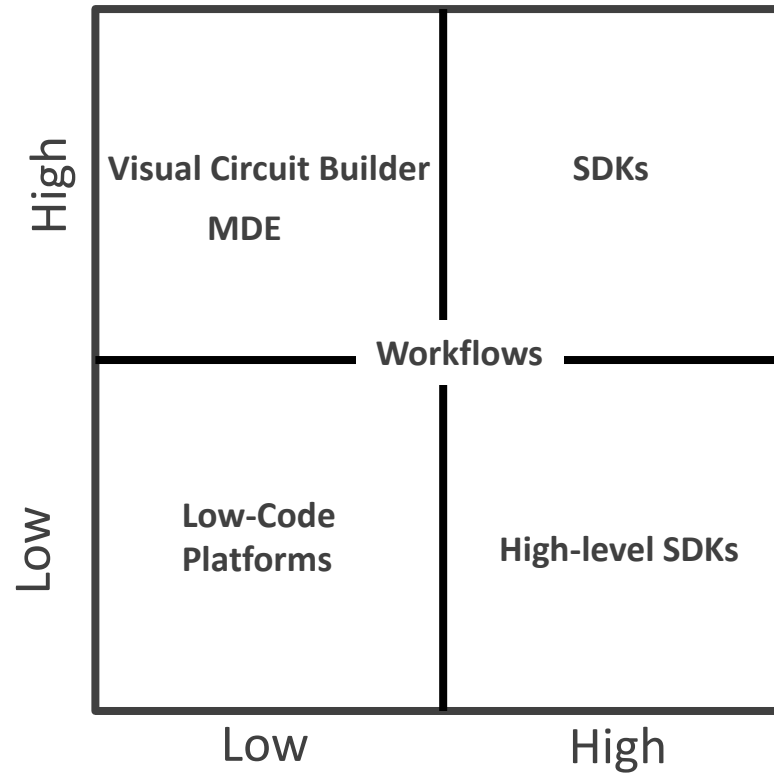
Lavinia Stiliadou

stiliadou@iaas.uni-stuttgart.de

Institute for Architecture of Application Systems

Motivation

Quantum Computing
Knowledge



Programming Knowledge

Motivation

- Existing quantum low-code platforms:
 - Often closed-source
 - Vendor-locked
 - Circuit-level only

Research Question

- To make quantum computing accessible:
 - Users need visual abstractions instead of low-level coding
 - Hybrid quantum-classical operations must be represented intuitively
 - Different expertise levels must be supported
 - Models should remain reusable and extensible

Research Question:

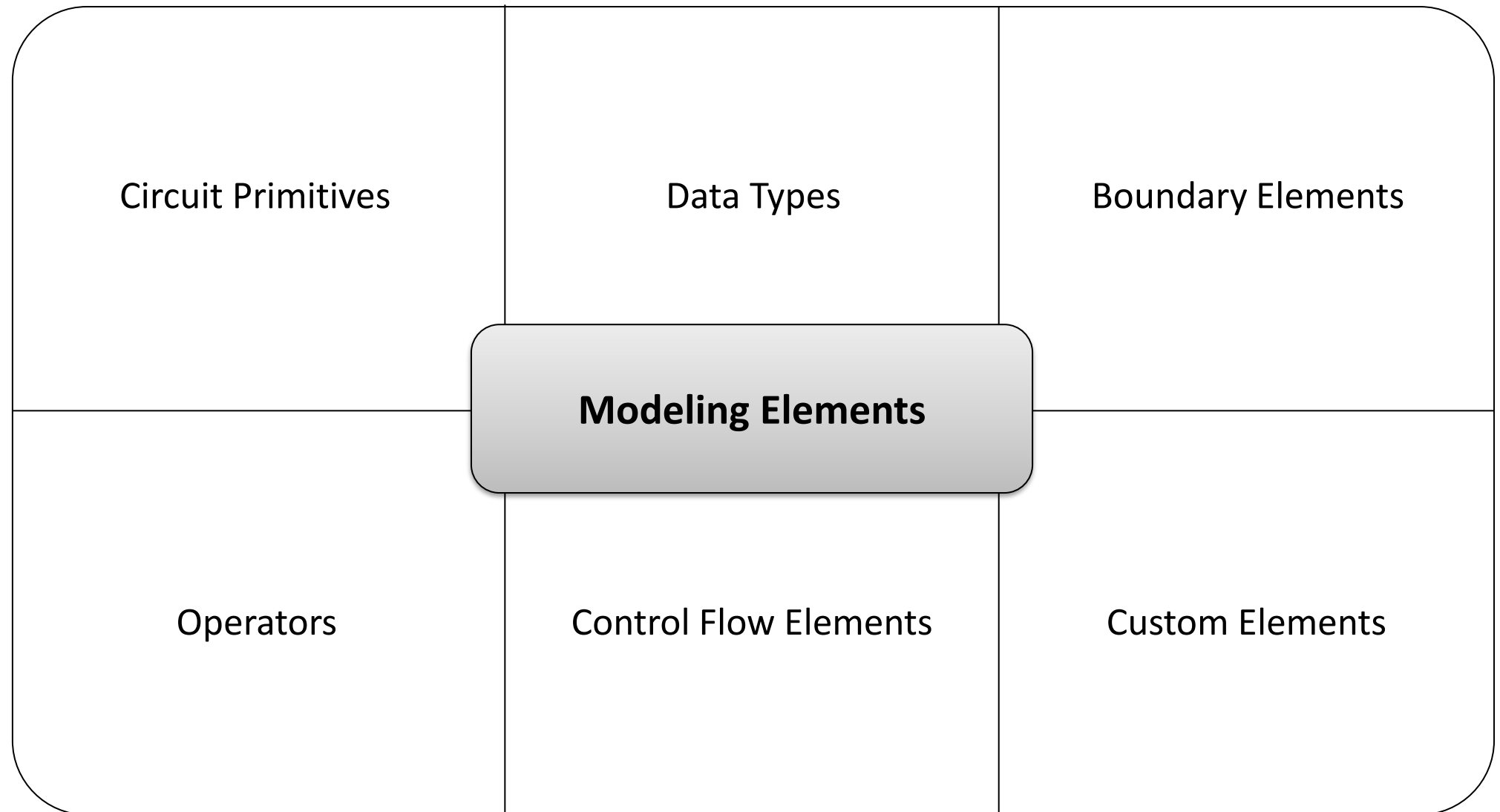
Which **abstractions** and **modeling elements** are required to support modeling of quantum algorithms in a **low-code platform**?

Modeling Language

- Icon
 - Basis Encoding Pattern Icon
- Type
 - State Preparation
- Properties
 - Encoding Type
 - Bound
 - Implementation (optional)
- Inputs
- Value
 - Value
- Outputs
 - Quantum Register

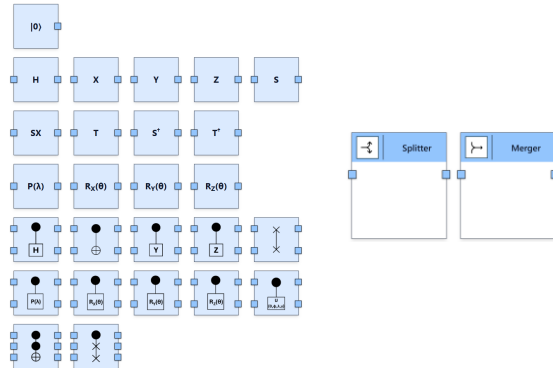
The diagram illustrates the 'Encode Value' block, which is a component in a modeling language. It features a blue header bar with the title 'Encode Value' and a small icon of a quantum circuit. Below the header, there are two main sections: 'Encoding Type' and 'Bound'. The 'Encoding Type' section has a dropdown menu currently set to 'Basis Encoding'. The 'Bound' section has a text input field with the placeholder 'Enter bound'. Below these sections is a yellow box labeled 'Value' with the text 'type: any'. To the right of the 'Value' box is a light blue box labeled 'Output:' with the text 'type: quantum register'. This box contains two input fields: 'Identifier' and 'Size'. The 'Identifier' field is a solid blue box, and the 'Size' field is a dashed blue box. A small blue square is located to the right of the 'Identifier' field.

Modeling Language



Modeling Language

Circuit Primitives



Data Types

Boundary Elements

Modeling Elements

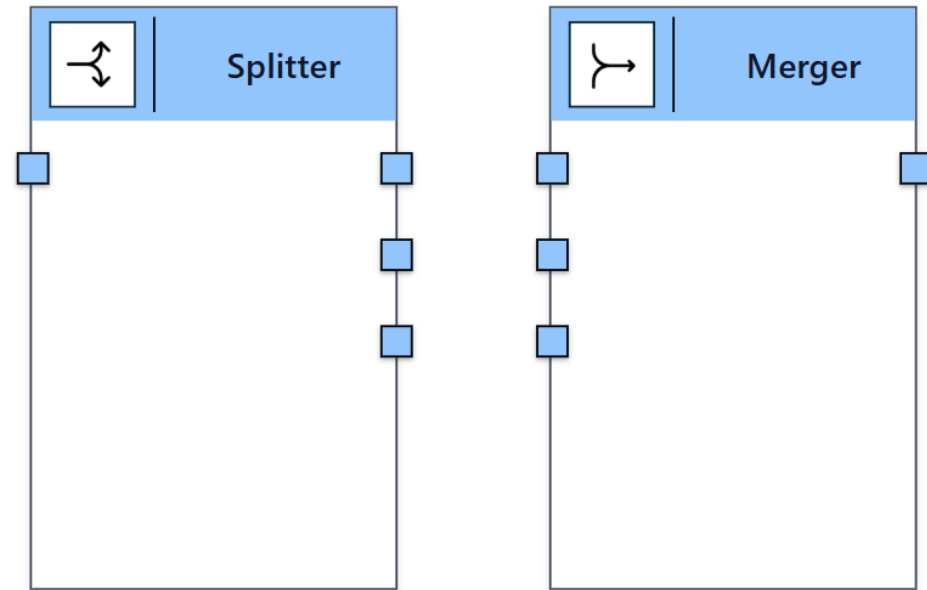
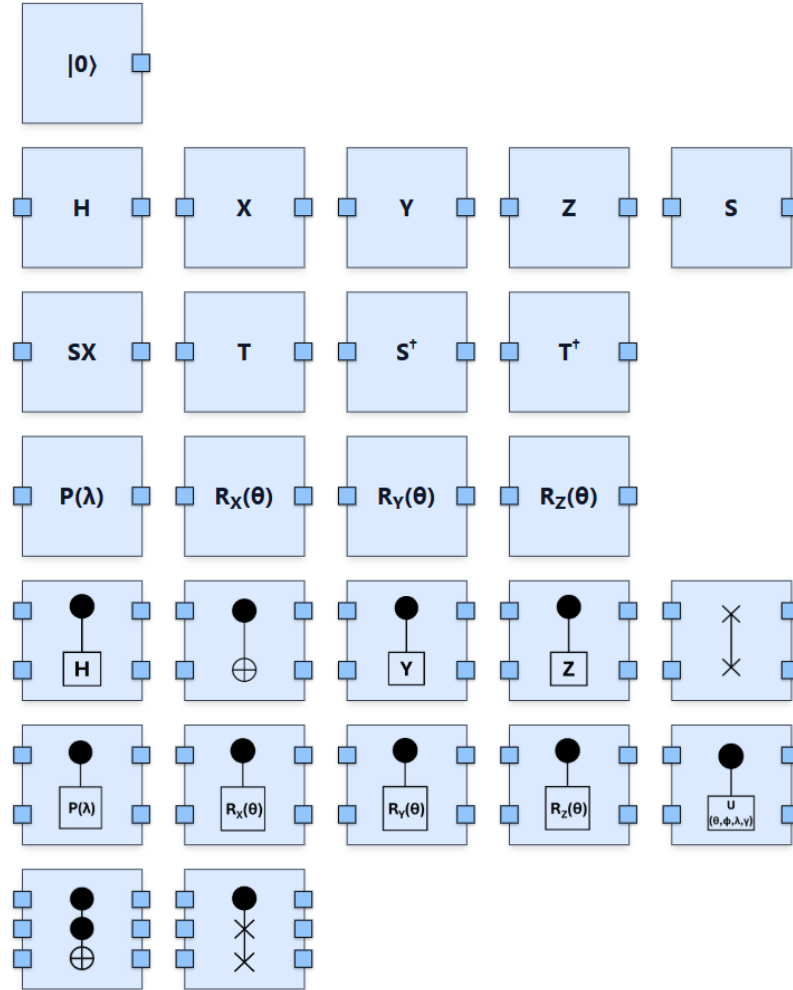
Operators

Control Flow Elements

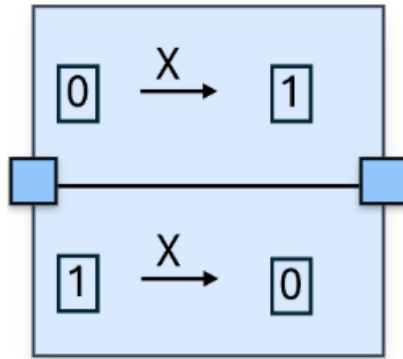
Custom Elements

Modeling Language

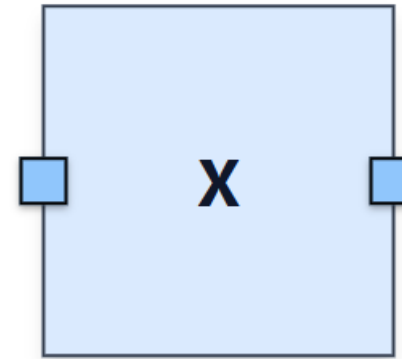
■ Circuit Primitives:



- **Circuit Primitives:**



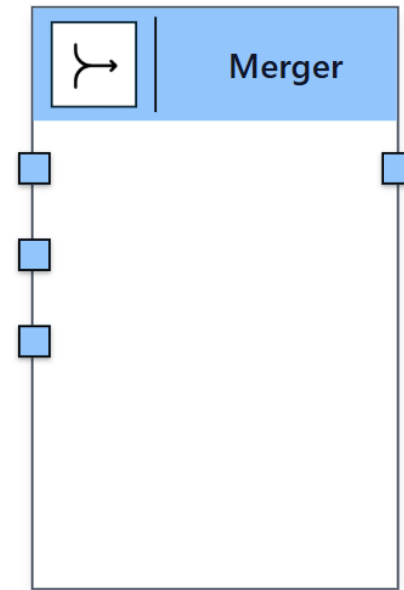
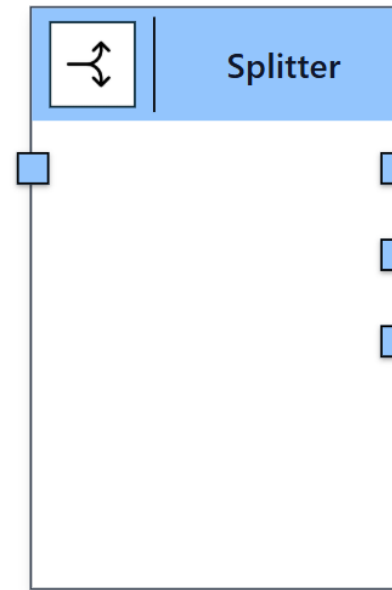
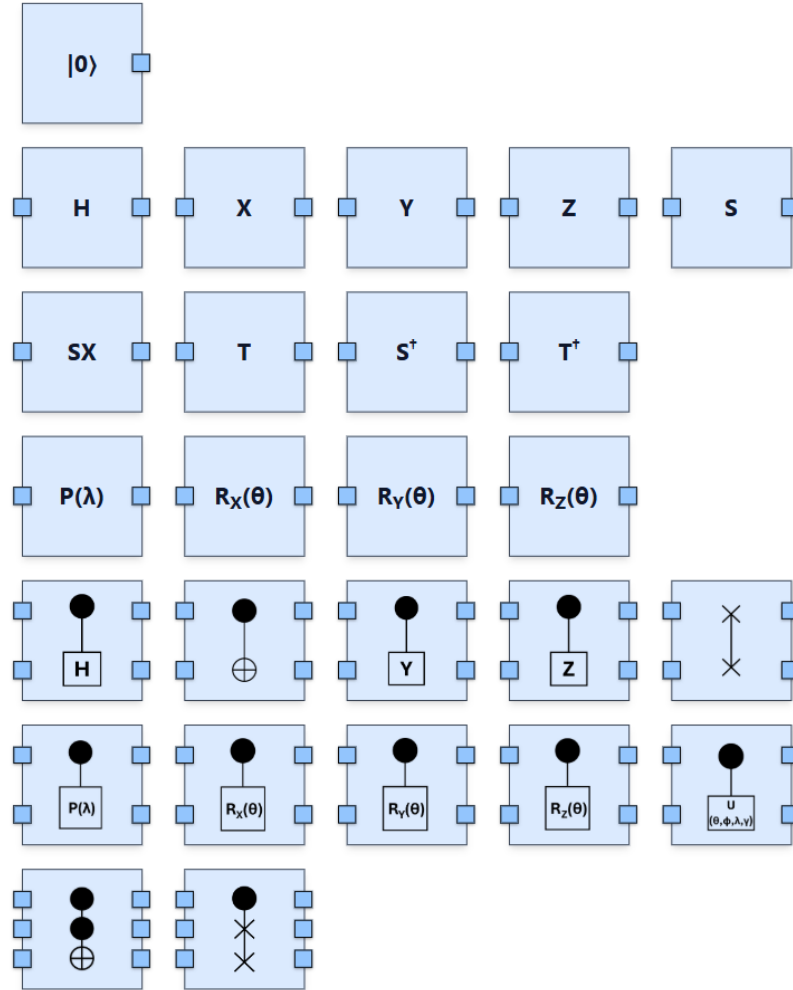
Explorer



Pioneer

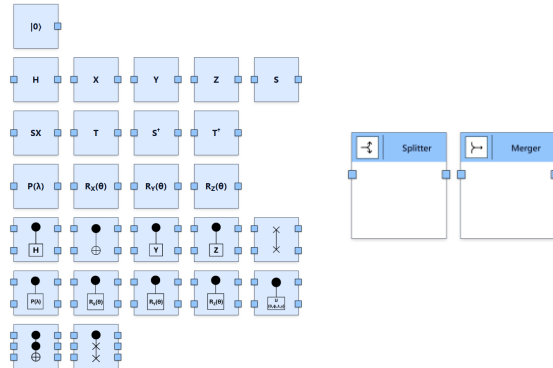
Modeling Language

■ Circuit Primitives:



Modeling Language

Circuit Primitives



Data Types

Boundary Elements

Modeling Elements

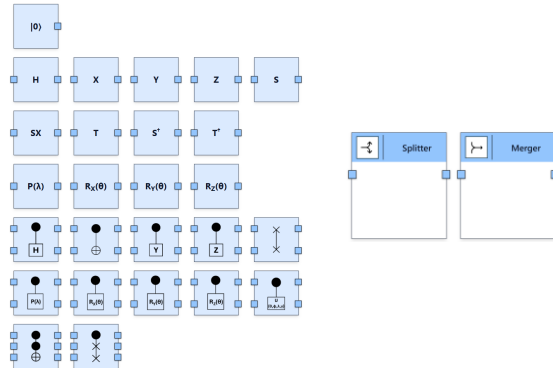
Operators

Control Flow Elements

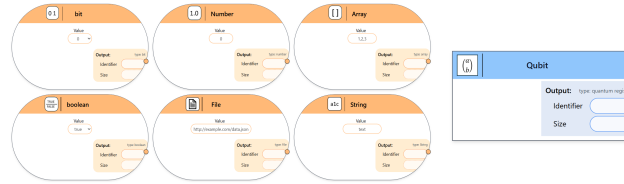
Custom Elements

Modeling Language

Circuit Primitives



Data Types



Boundary Elements

Modeling Elements

Operators

Control Flow Elements

Custom Elements

Modeling Language

■ Data Types:

0 1

bit

Value
0 ▾

Output: type: bit
Identifier
Size

1.0

Number

Value
0

Output: type: number
Identifier
Size

[]

Array

Value
1,2,3

Output: type: array
Identifier
Size

TRUE
FALSE

boolean

Value
true ▾

Output: type: boolean
Identifier
Size



File

Value
<http://example.com/data.json>

Output: type: File
Identifier
Size

a1c

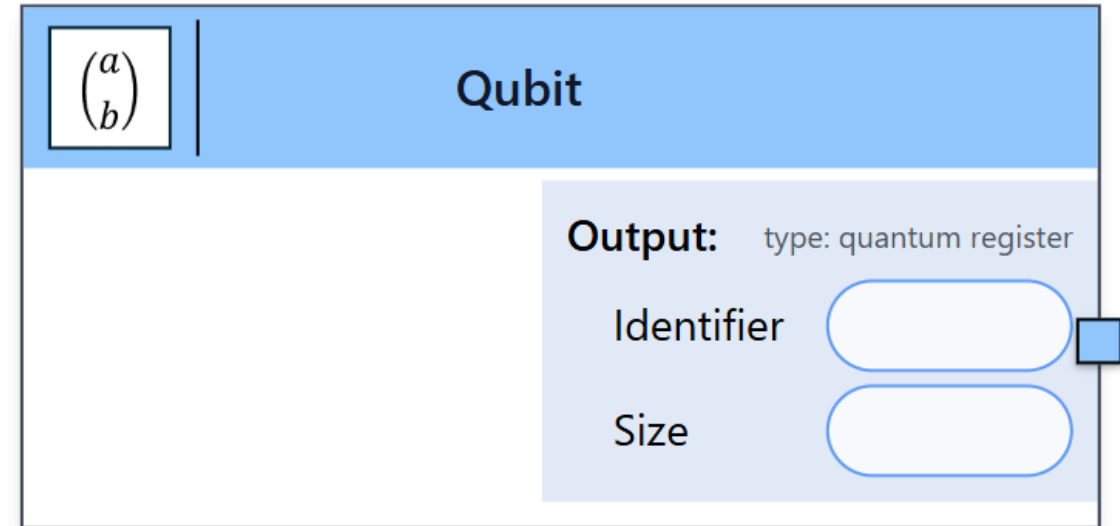
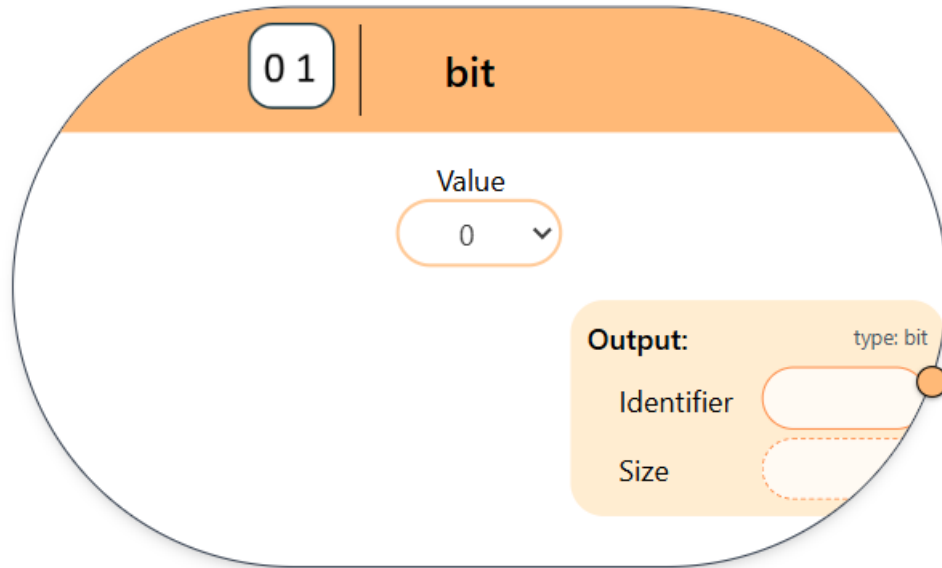
String

Value
text

Output: type: String
Identifier
Size

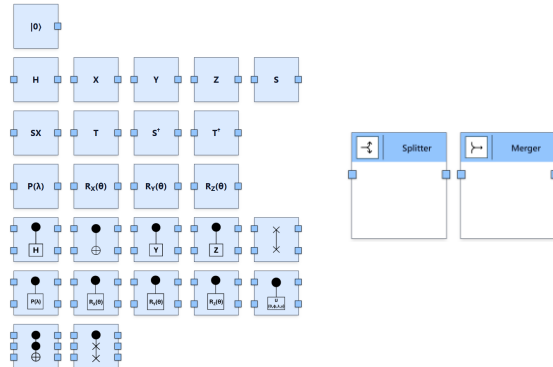
Modeling Language

- Data Types:

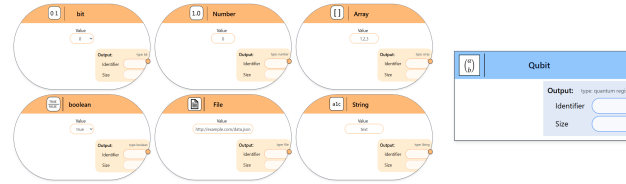


Modeling Language

Circuit Primitives



Data Types



Boundary Elements

Modeling Elements

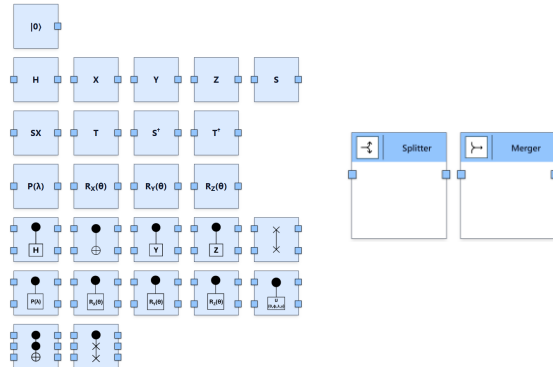
Operators

Control Flow Elements

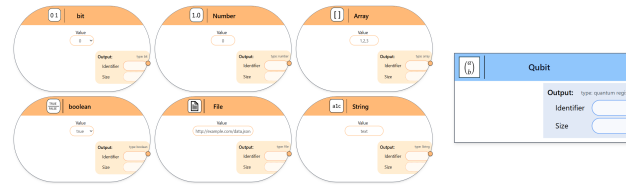
Custom Elements

Modeling Language

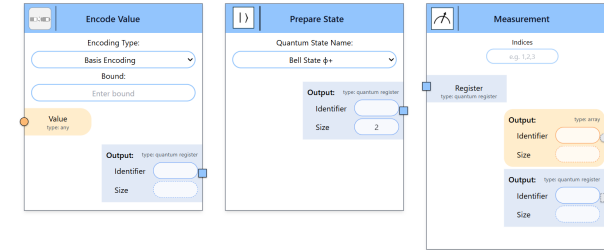
Circuit Primitives



Data Types



Boundary Elements



Modeling Elements

Operators

Control Flow Elements

Custom Elements

Modeling Language

■ Boundary Elements:

 **Encode Value**

Encoding Type:

Basis Encoding

Bound:

Enter bound

Value

type: any

Output: type: quantum register

Identifier

Size

 **Prepare State**

Quantum State Name:

Bell State $\phi+$

Output: type: quantum register

Identifier

Size

2

 **Measurement**

Indices

e.g. 1,2,3

Register

type: quantum register

Output: type: array

Identifier

Size

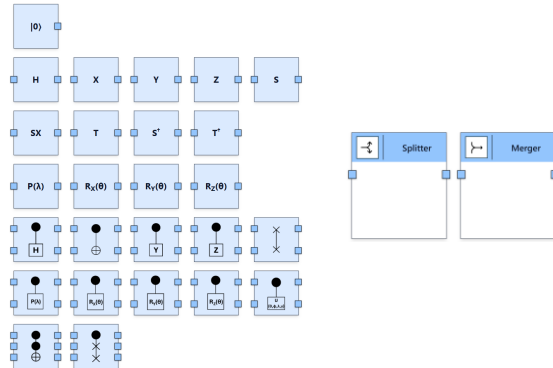
Output: type: quantum register

Identifier

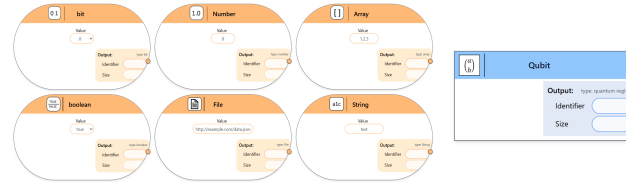
Size

Modeling Language

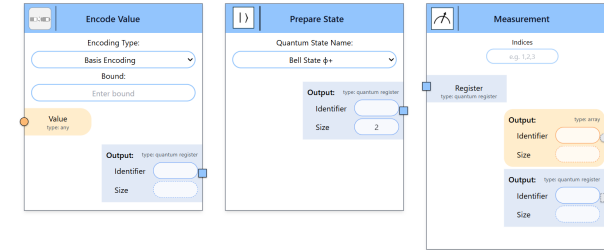
Circuit Primitives



Data Types



Boundary Elements



Modeling Elements

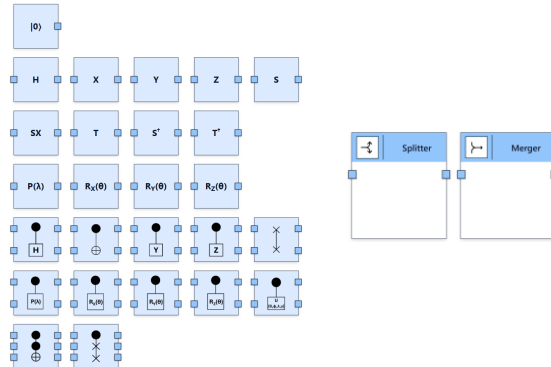
Operators

Control Flow Elements

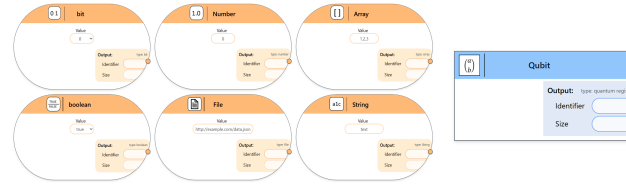
Custom Elements

Modeling Language

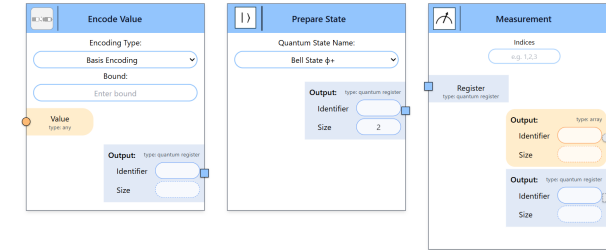
Circuit Primitives



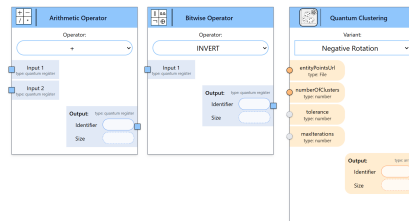
Data Types



Boundary Elements



Modeling Elements



Operators

Control Flow Elements

Custom Elements

■ Operatoren:

+

-

/

*

Arithmetic Operator

Operator:

+

Input 1

type: quantum register

Input 2

type: quantum register

Output:

type: quantum register

Identifier

Size

||

&&

¬

⊕

Bitwise Operator

Operator:

INVERT

Input 1

type: quantum register

Output:

type: quantum register

Identifier

Size

Quantum Clustering

Variant:

Negative Rotation

entityPointsUrl

type: File

numberOfClusters

type: number

tolerance

type: number

maxIterations

type: number

Output:

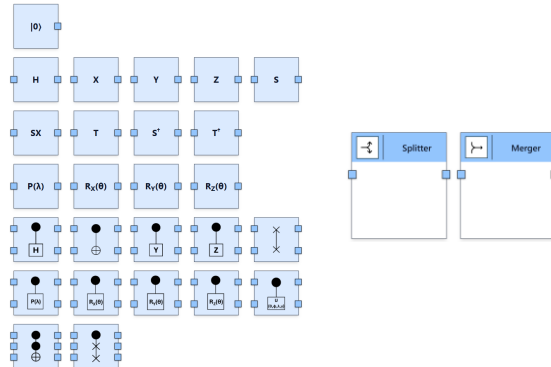
type: array

Identifier

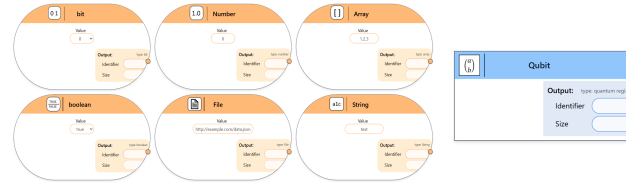
Size

Modeling Language

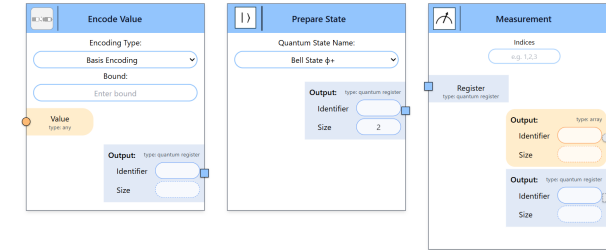
Circuit Primitives



Data Types

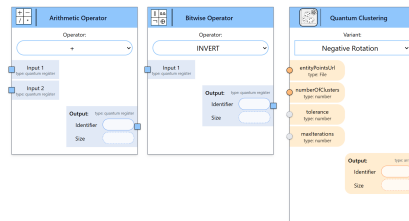


Boundary Elements



Modeling Elements

Operators

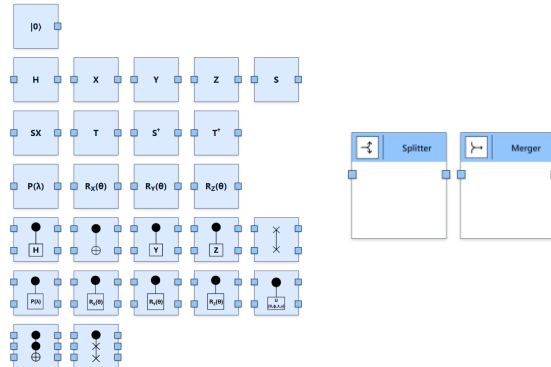


Control Flow Elements

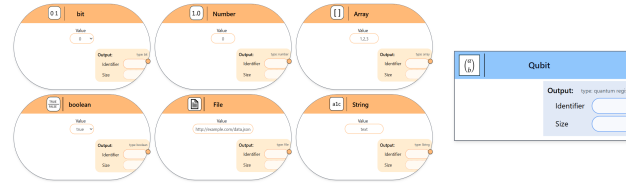
Custom Elements

Modeling Language

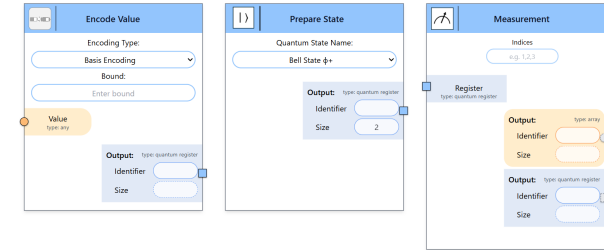
Circuit Primitives



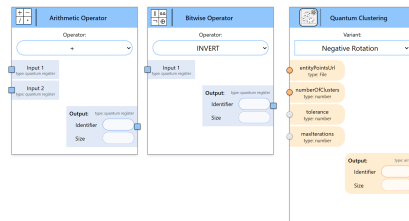
Data Types



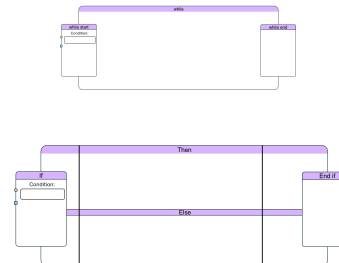
Boundary Elements



Modeling Elements



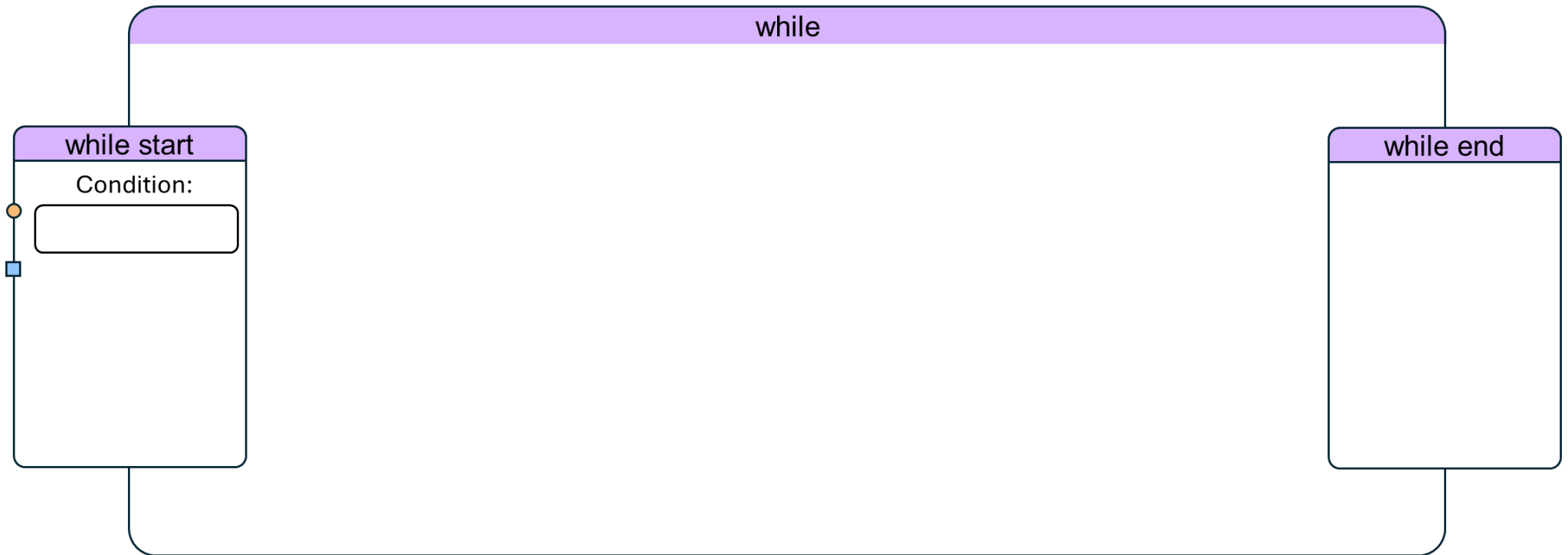
Operators



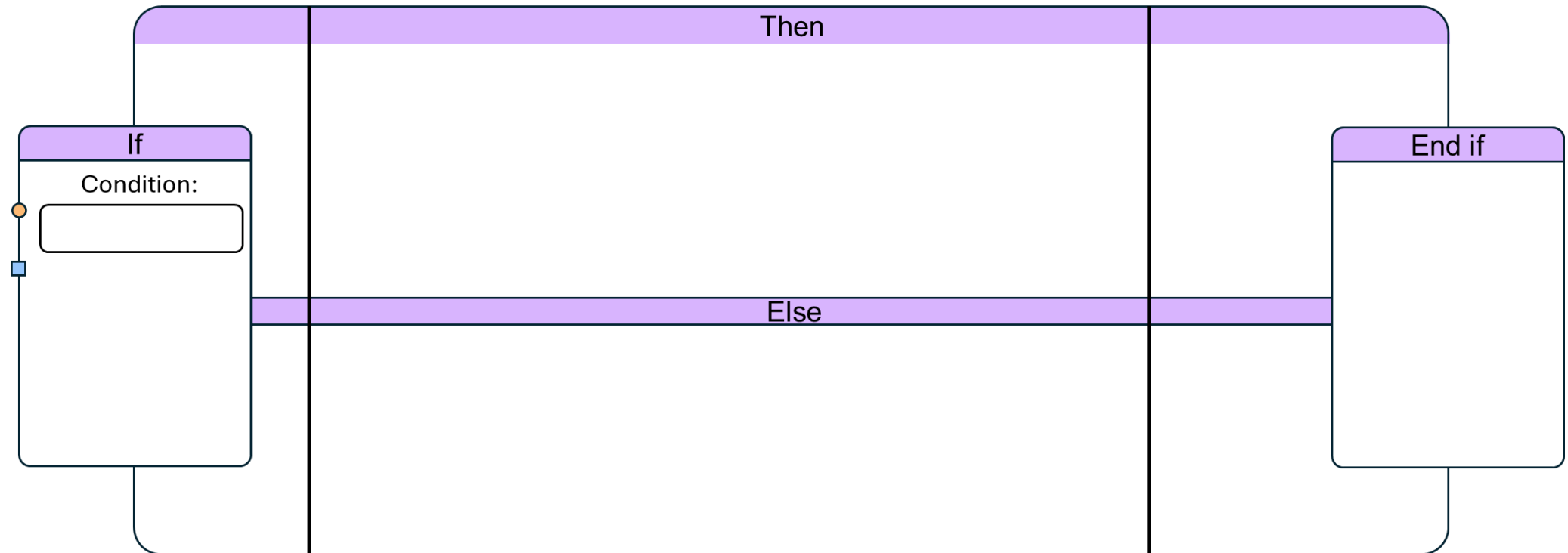
Control Flow Elements

Custom Elements

- Control Flow Elements:

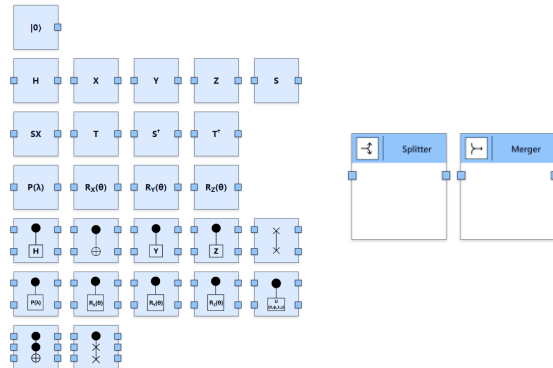


- Control Flow Elements:

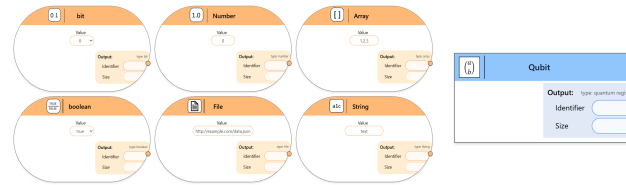


Modeling Language

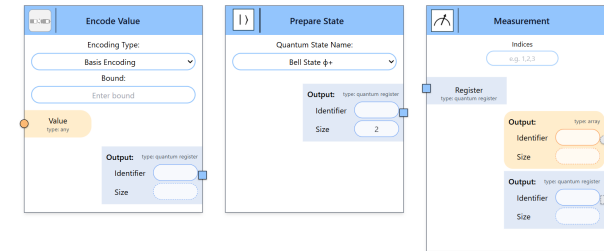
Circuit Primitives



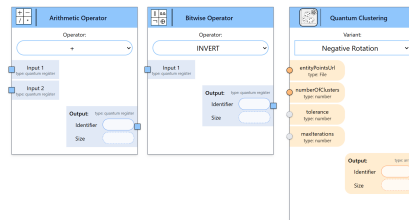
Data Types



Boundary Elements

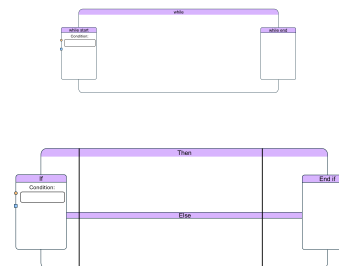


Modeling Elements

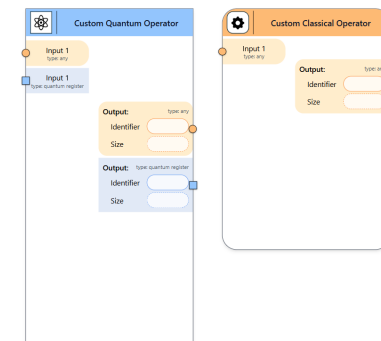


Operators

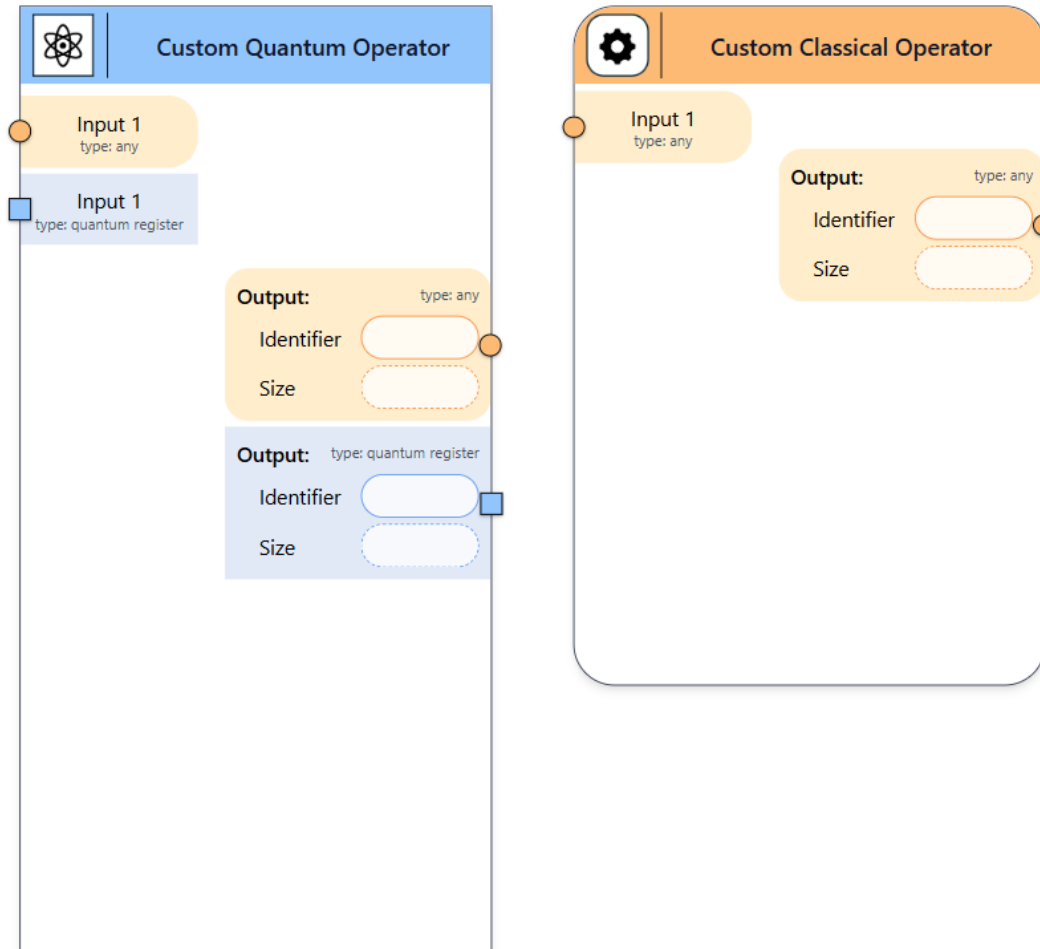
Control Flow Elements



Custom Elements

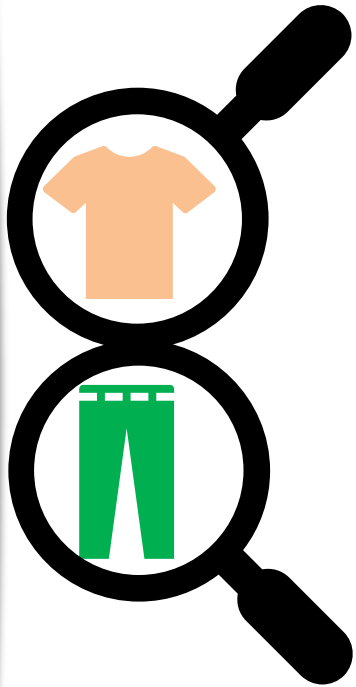


■ Custom Elements:



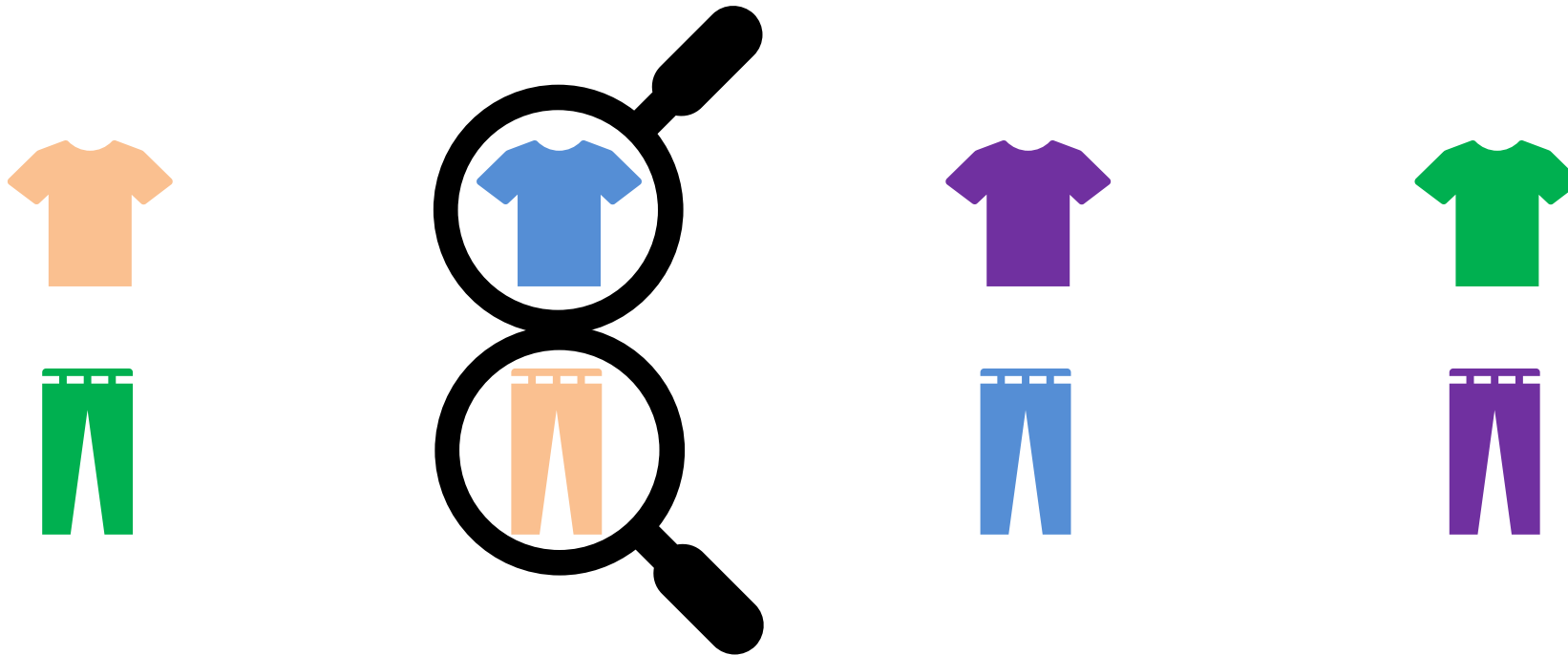
Task for Today

- Classical Solution:



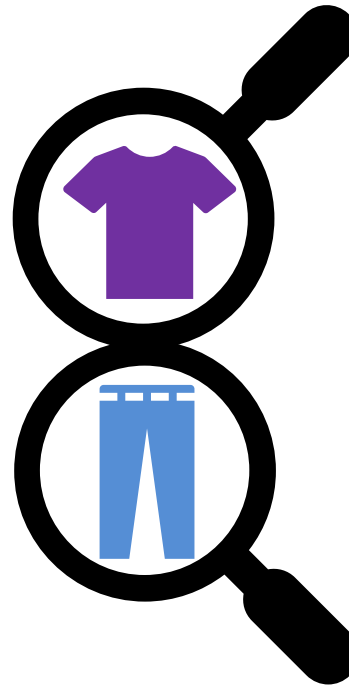
Task for Today

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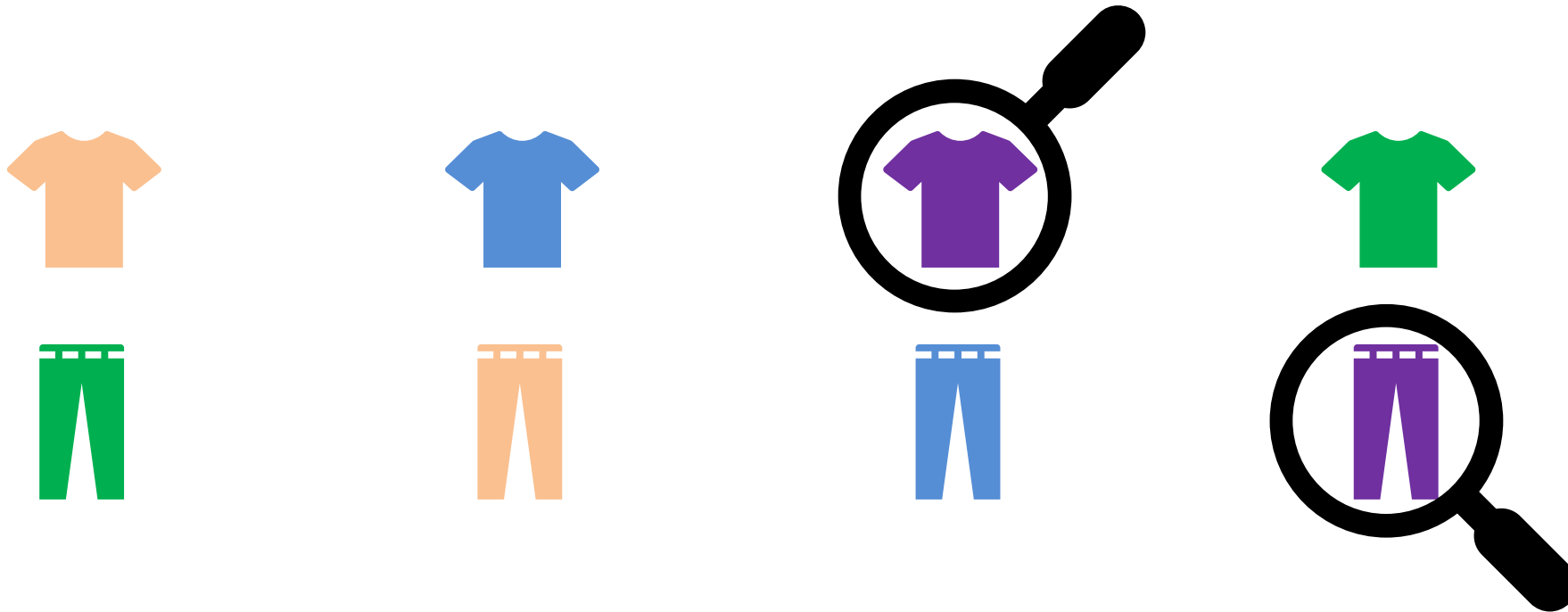
Task for Today

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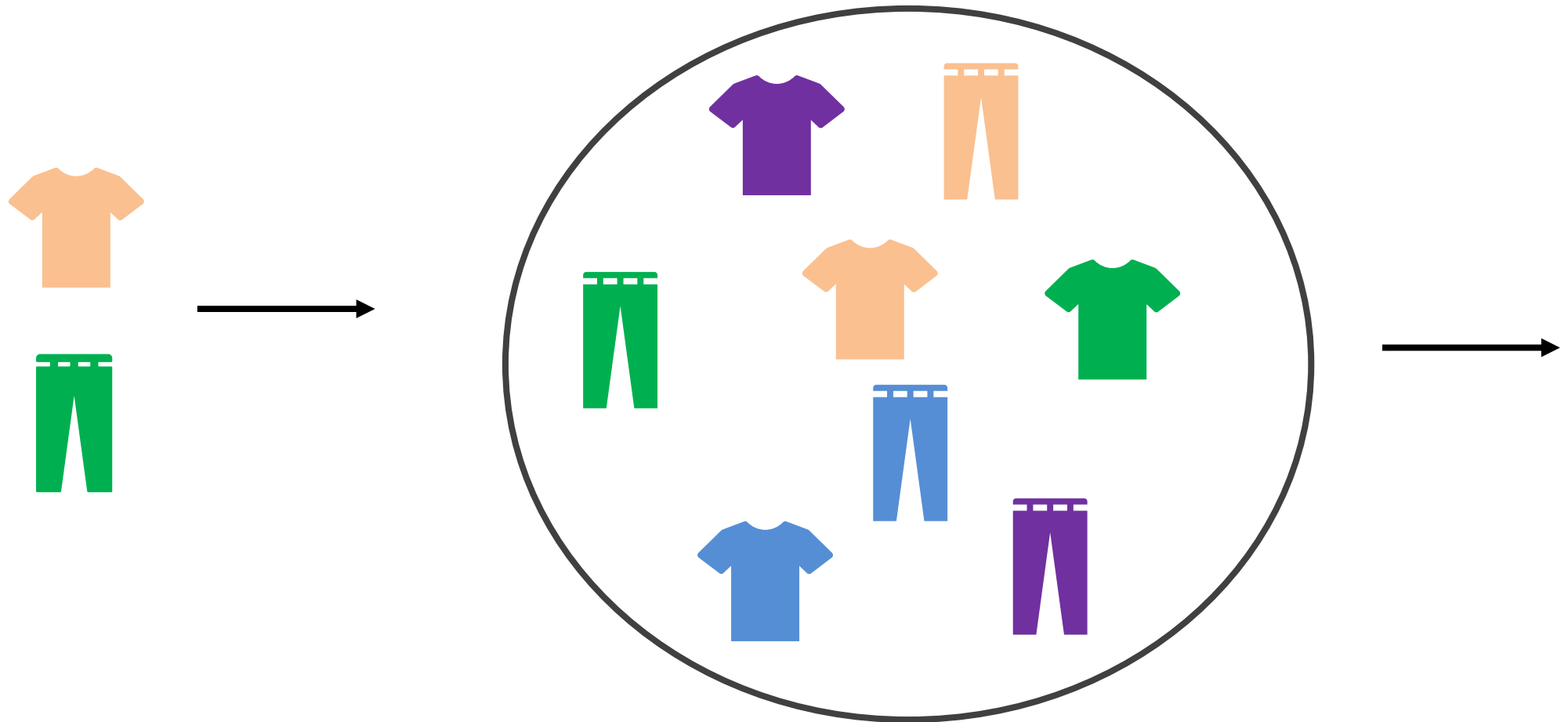
Task for Today

- Classical Solution:



Task for Today

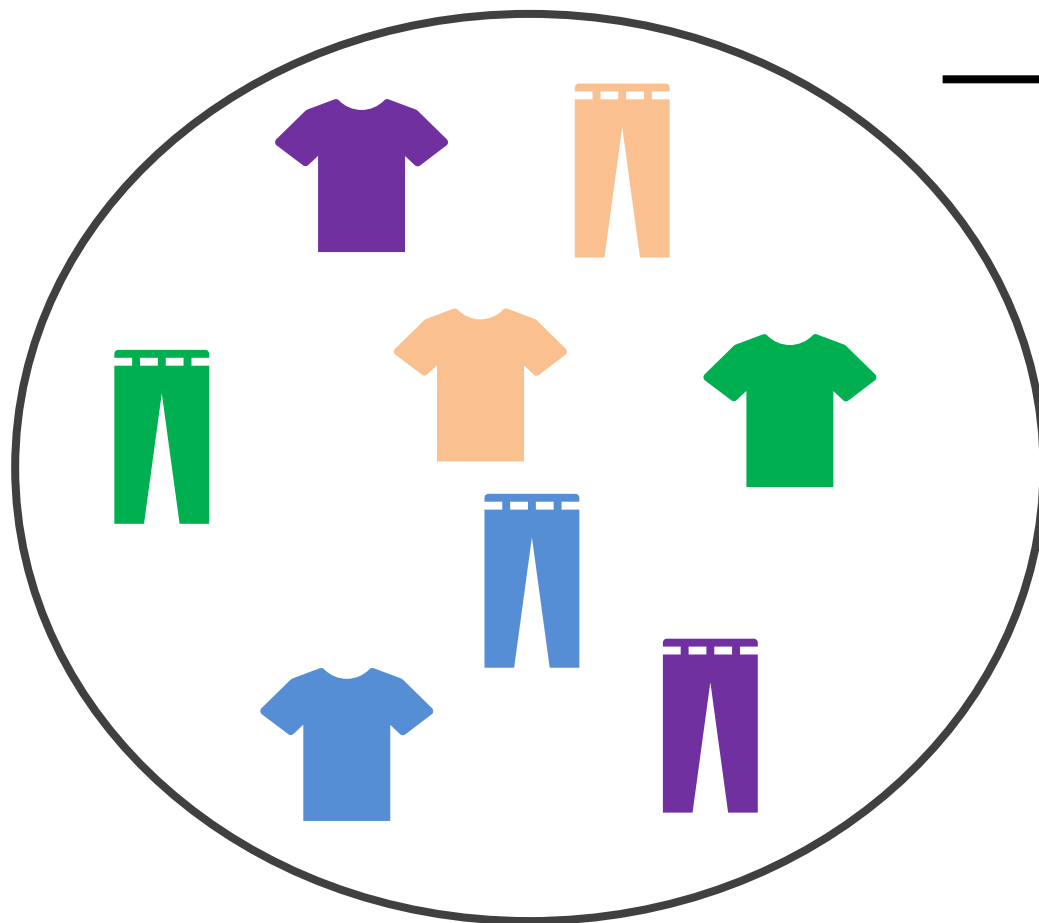
- Quantum Solution:



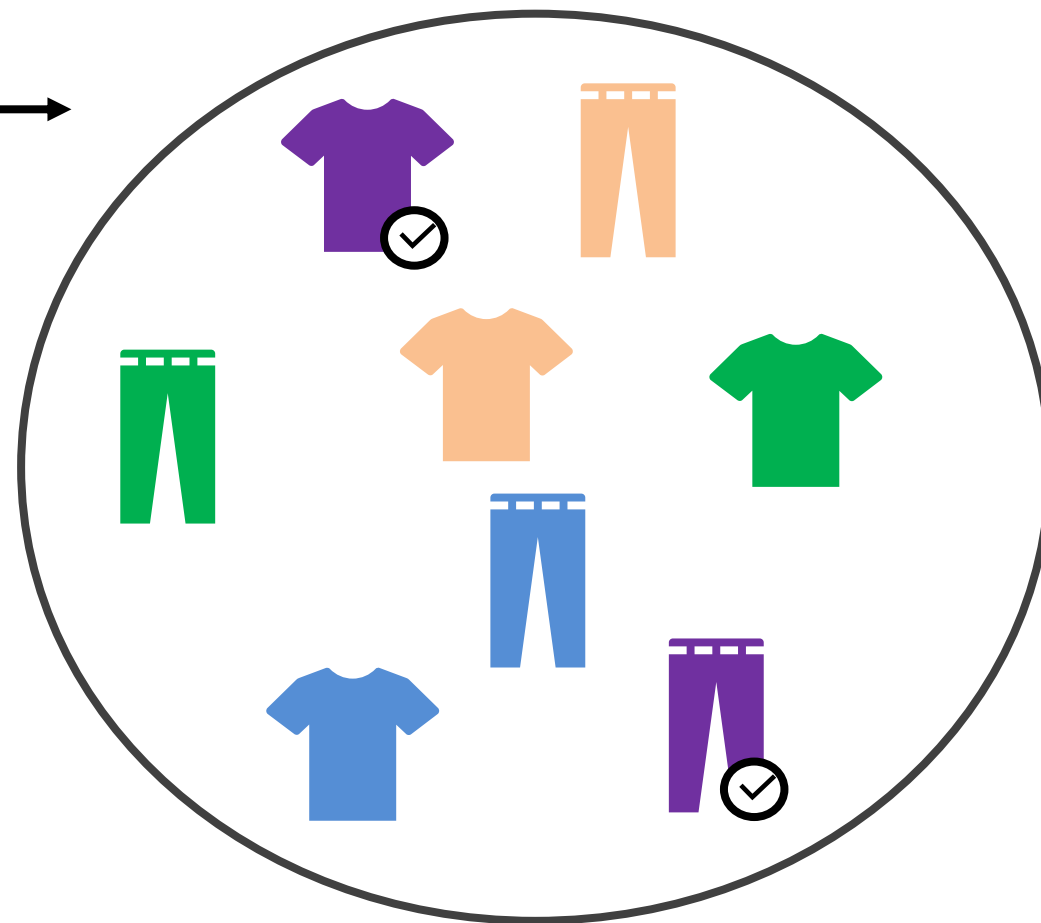
Uniform Superposition

Task for Today

- Quantum Solution:



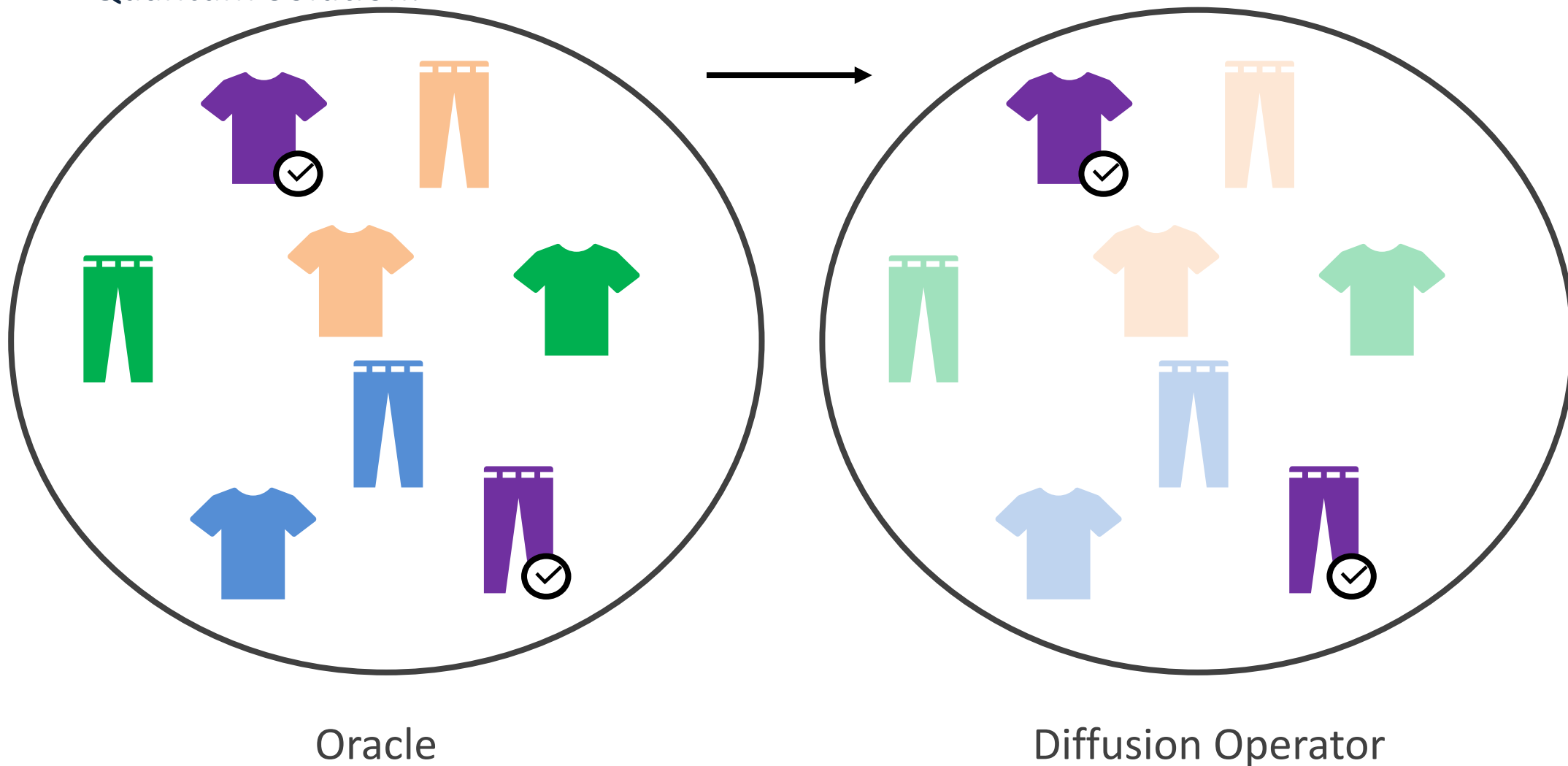
Uniform Superposition



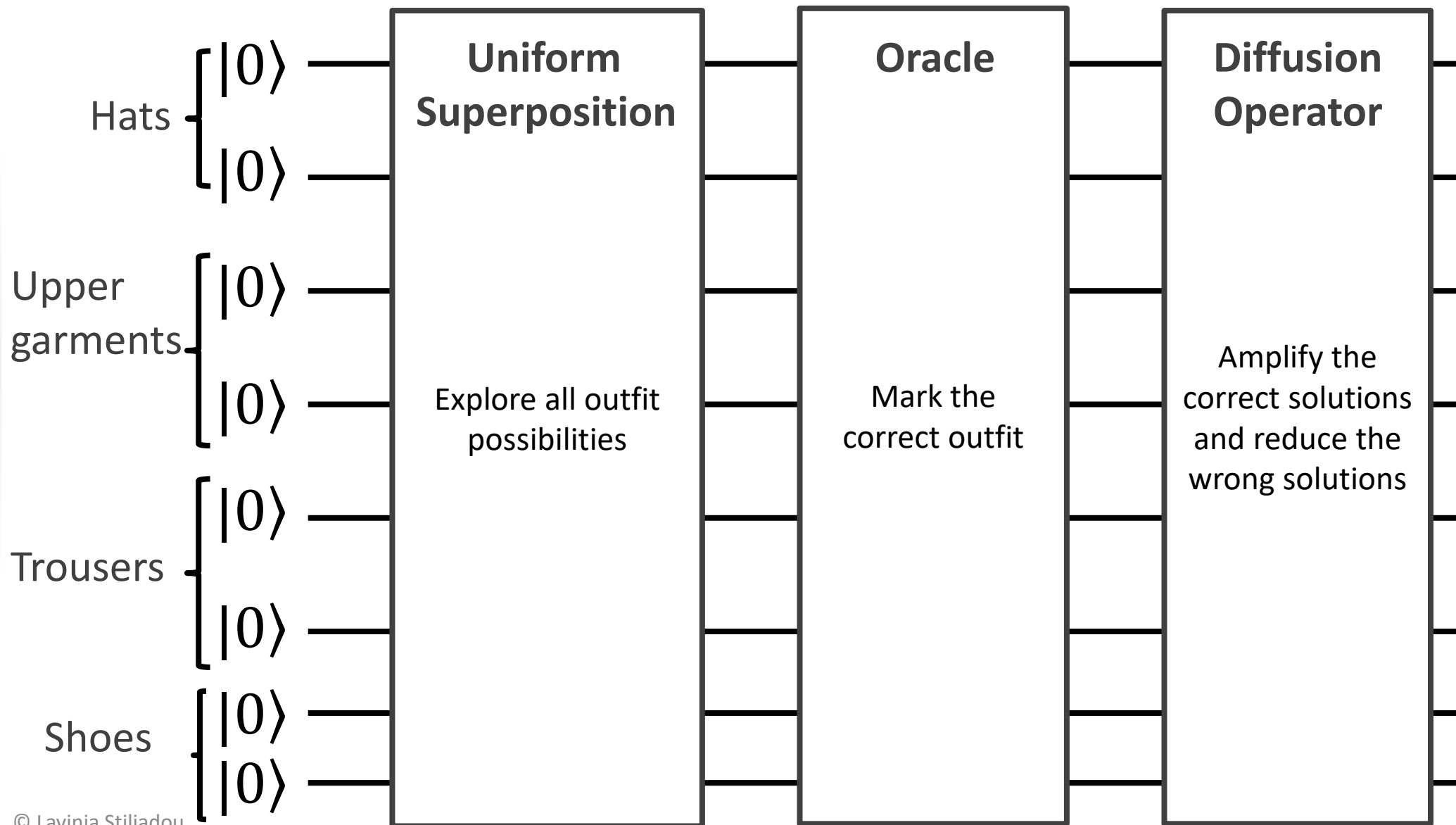
Oracle

Task for Today

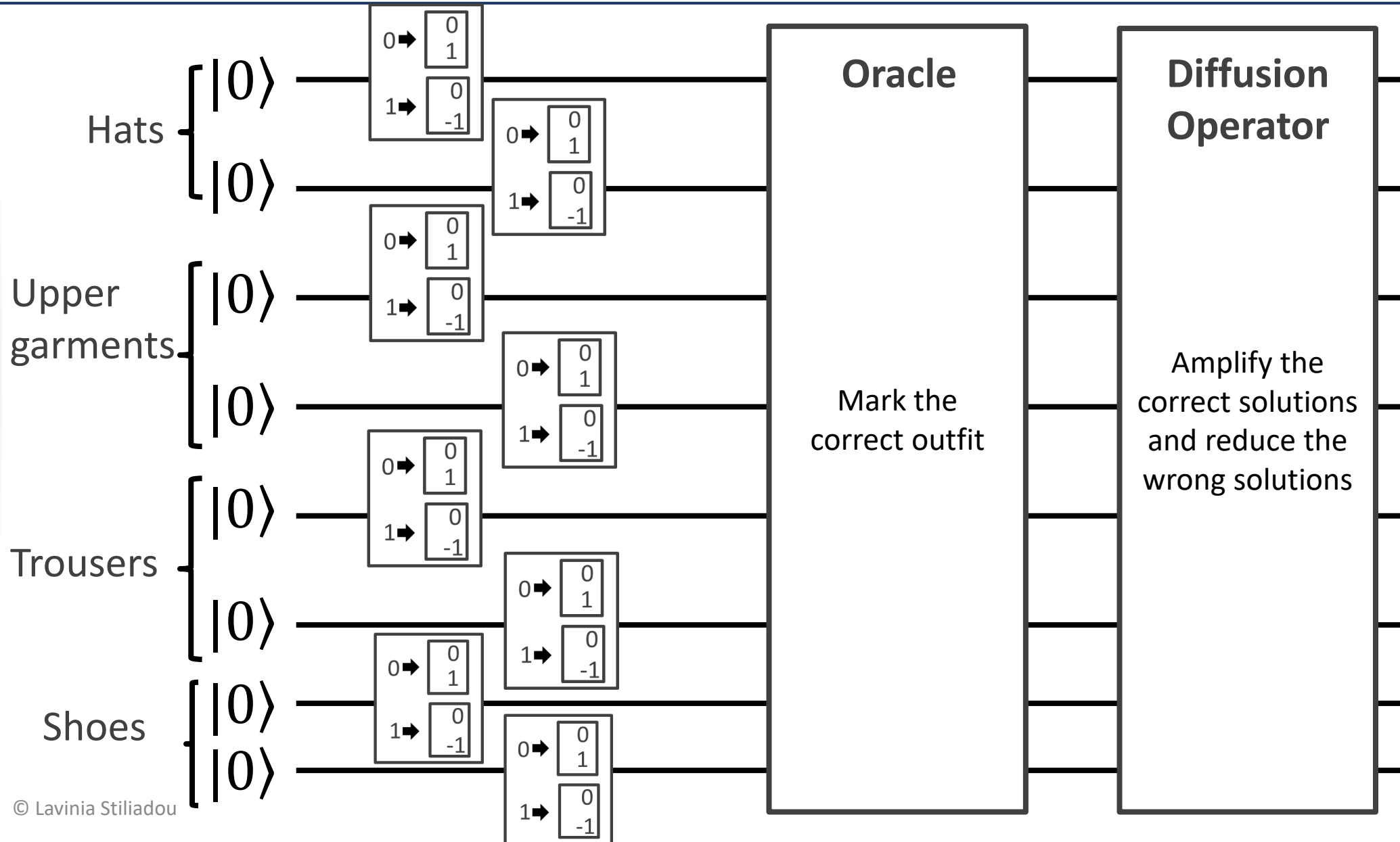
- Quantum Solution:



Task for Today - Grover's Search Algorithm



Task for Today - Grover's Search Algorithm

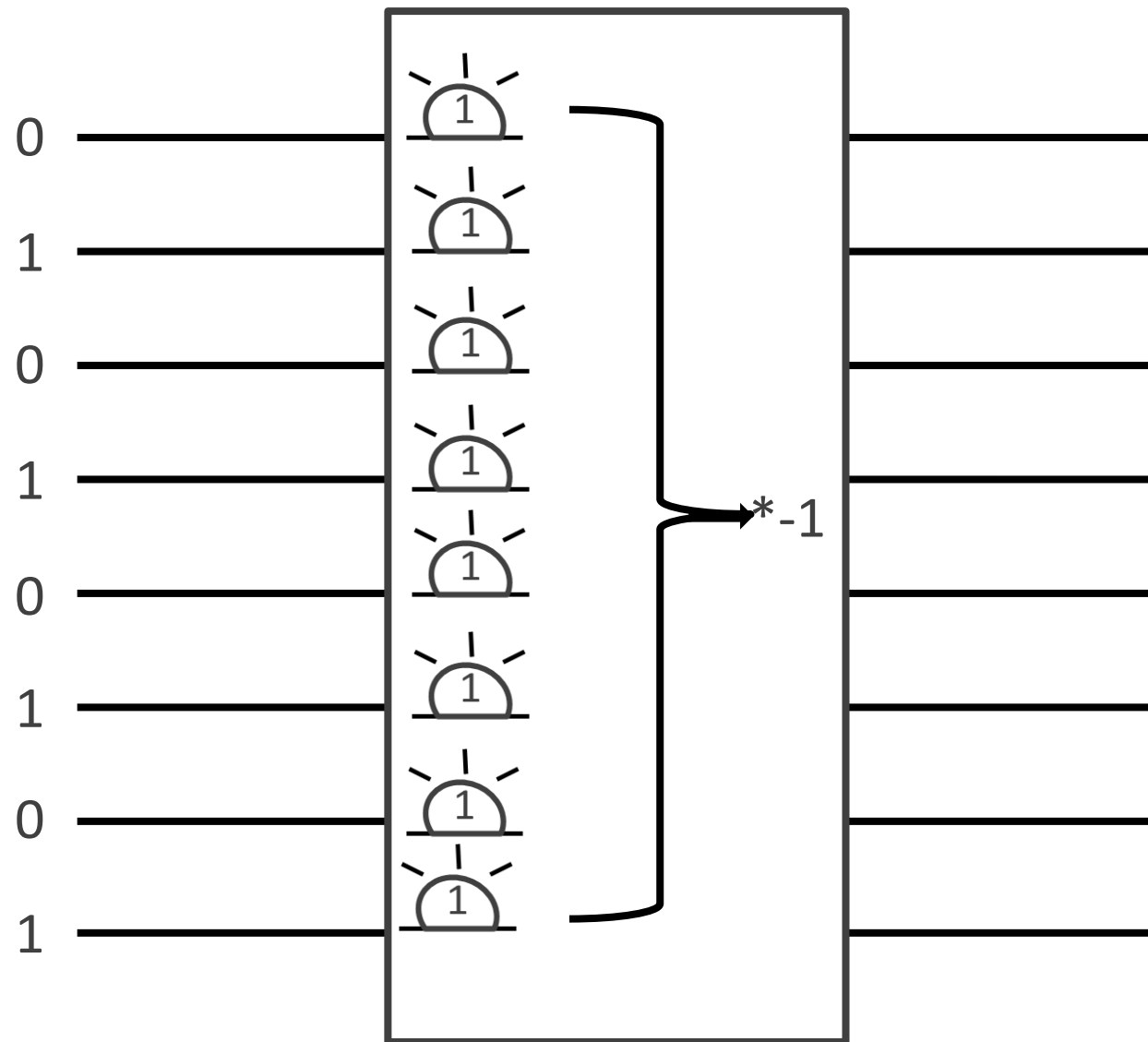


Task for Today - Grover's Search Algorithm Oracle

- The oracle should mark the correct solution
- We have 8 qubits
- If we want always the second option:
 - Second hat $\rightarrow$ 01
 - Second upper garment $\rightarrow$ 01
 - Second trouser $\rightarrow$ 01
 - Second shoe $\rightarrow$ 01

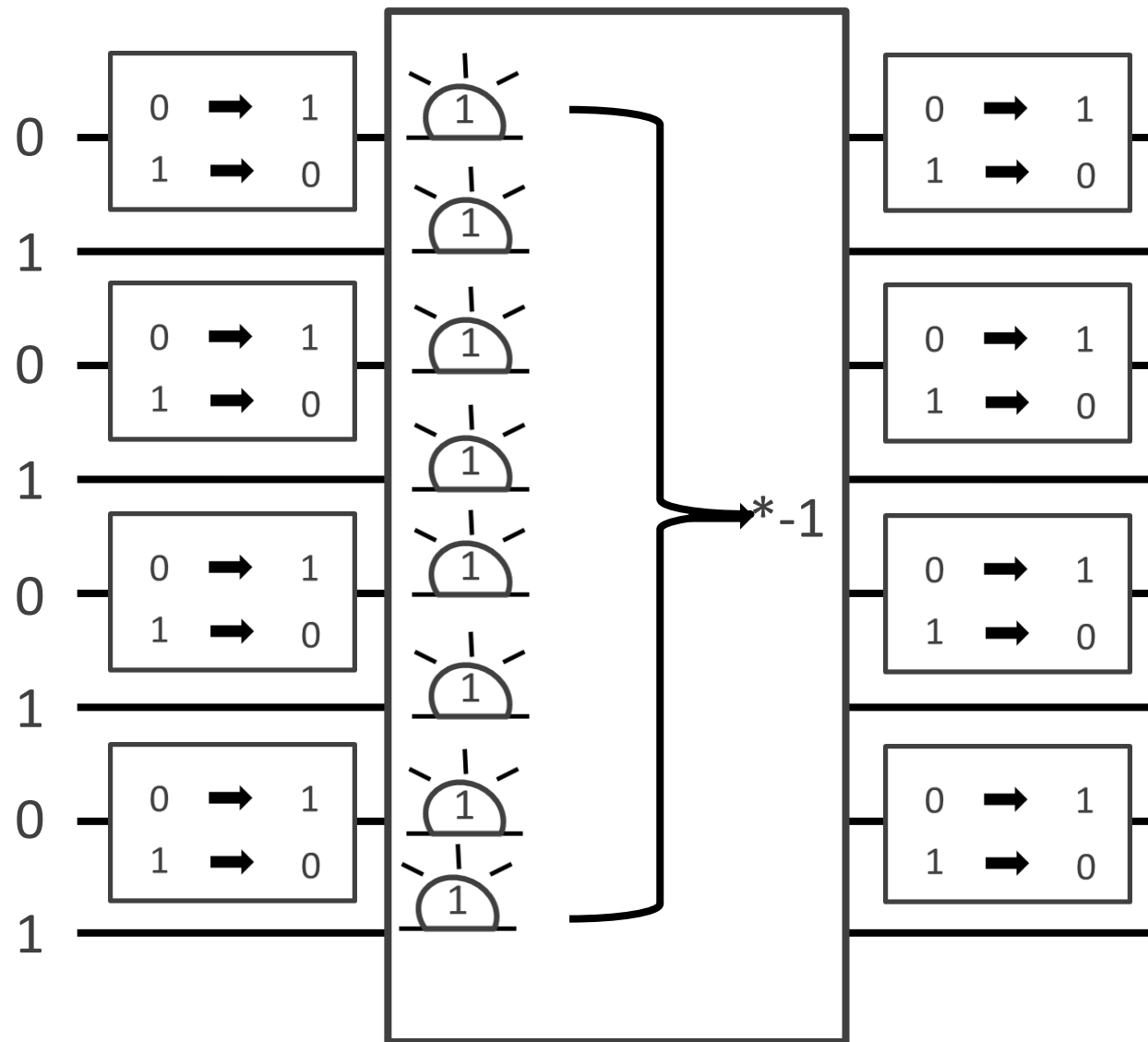
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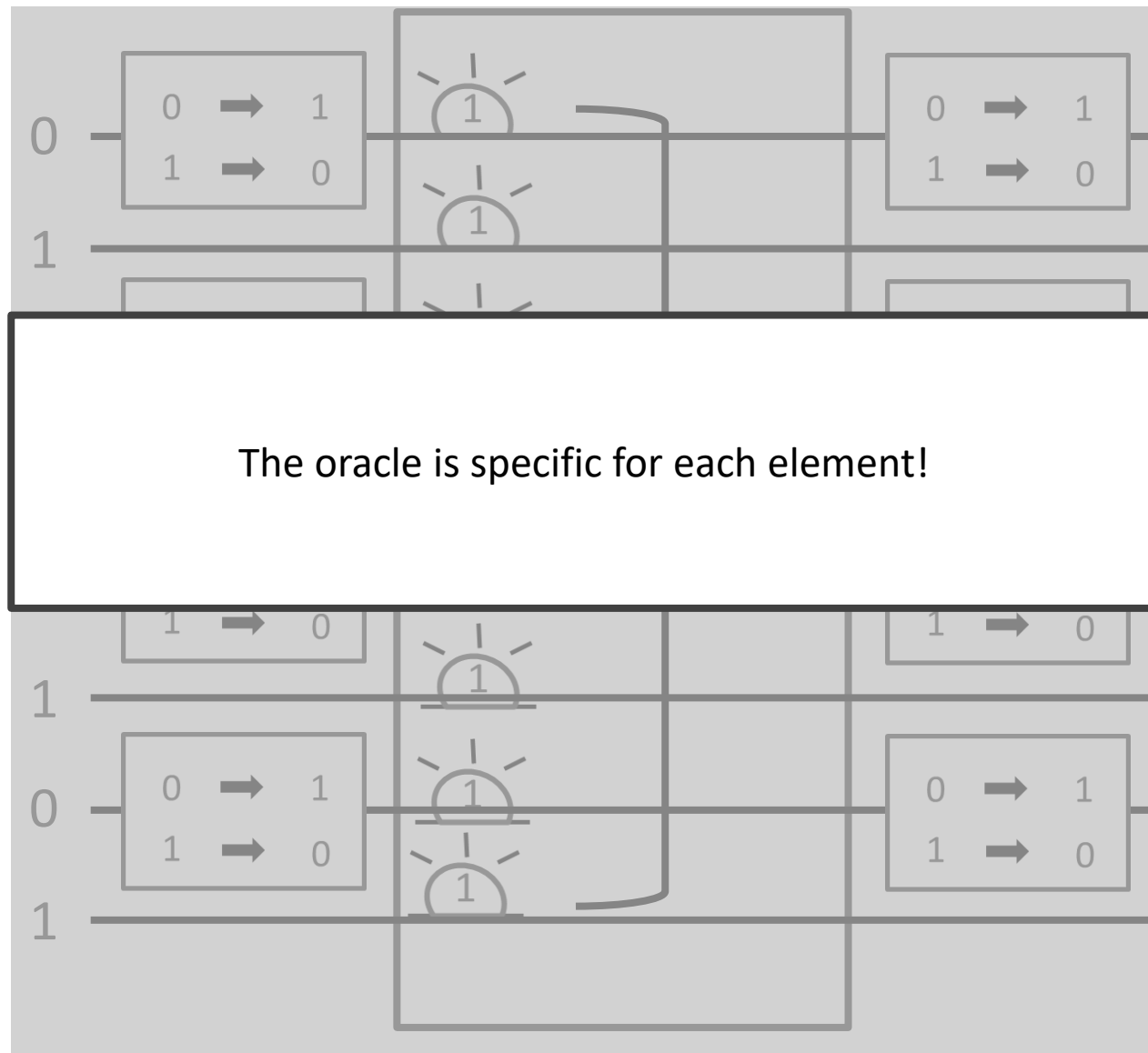
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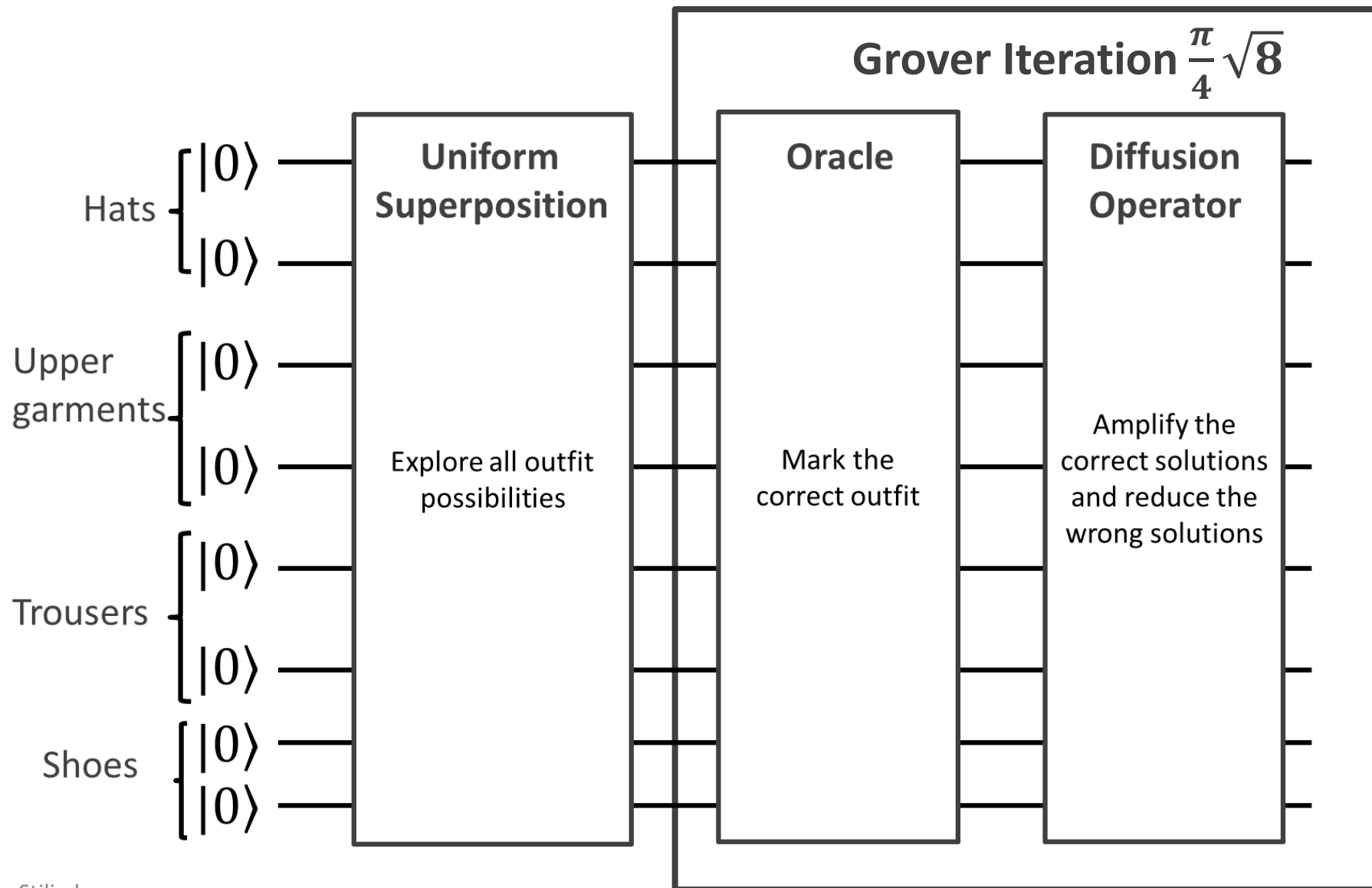


Task for Today - Grover's Search Algorithm Oracle

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Task for Today - Grover's Search Algorithm



Qonstruct Overview

The screenshot shows the Qonstruct application interface. At the top is a toolbar (1) with icons for file operations, templates, configuration, domain profiles, and help. On the left is a palette (2) with categories: Boundary Elements, Circuit Primitives, Data Types, Operators, Custom Operators, and Machine Learning (ML) Elements. In the center is the Experience Mode Panel (3) with settings for Domain Profile, Experience Level, Ancilla Modeling, Compact Visualization, Completion Guaranteed, and Collaboration Mode, along with a Recommended Settings table. The main area is the Canvas (4), a grid for building circuits. At the bottom right is a Mini Map (5) showing a circuit diagram. On the right is the Properties Provider (6) with fields for Version, Name, Author, Description, and Optimization options.

Experience Mode Panel (3)

Domain Profile	None
Experience Level	Explorer
Ancilla Modeling	Off
Compact Visualization	Off
Completion Guaranteed	Yes
Collaboration Mode	Off

Recommended Settings

Experience Level	Ancilla Mode	Compact Visualization
Explorer	Off	Off
Pioneer	On	On

Properties Provider (6)

Version: 1.0.0

Name: My Model

Author:

Description: This is a model.

Optimization

- ☐ Enable Width Optimization
- ☐ Enable Depth Optimization

Legend:

- ① Toolbar ② Palette ③ Experience Mode Panel ④ Canvas ⑤ Mini Map ⑥ Properties Provider

Thank you for your attention!